Seat No _____

18 OCT 2021

BCA Semester – 3 Examination

Paper Name: Operating System | Paper Code: (22691)

Time duration: 2 Hours | Total Marks: 56

Que - 1 Define Operating system. Write a detail note on evolution of Operating System. [14]

OR

Que – 1

(A) Define functions of operating system. [7]

(B) Write a detail note on Client/Server architecture. [7]

Que – 2 Apply SJF (Preemptive), Priority (Preemptive), and Round Robin (Time-slice=5 milisec) process scheduling algorithm on the following process group and find out AWT and ATT and conclude which algorithm is best. [14]

Process	Arrival Time	Execution Time
A	00	10
B	13	23
C	05	12
D	21	20
E	30	15

OR

Que – 2

(A) Define process, also draw and explain process state diagram and queuing diagram [7]

(B) Define following terms: process, cpu utilization, throughput, turnaround time, waiting time, response time. [7]

Que – 3 Explain necessary conditions to occur deadlock and how to avoid deadlock [14]

OR

Que – 3

(A) Explain deadlock detection and recovery. [7]

(B) Define thread. Explain benefits of threads [7]

Que – 4 Define page fault and page replacement? Explain all page replacement algorithm with suitable example. [14]

OR

Que – 4

(A) Explain logical and physical address map [7]

(B) Write a detail note on continuous memory allocation technique. [7]